Spatial Regression Analysis of Income Inequality in NYC

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Spatial Econometrics and Data

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Introduction

New York City is one of the urban areas that faces an issue of income inequality. The Gini index is one of the measurements to understand this inequality, and its variation across neighborhoods can be influenced by factors such as education, income levels, public assistance, and unemployment. This analysis focuses on identifying the determinants of the Gini index at the census tract level, using socio-demographic variables to understand patterns of disparity across the city better.

Ordinary Least Squares (OLS) regression is a method often used to estimate relationships between dependent and independent variables. It assumes that the residuals are independent and identically distributed. This assumption usually fails in spatial data due to spatial autocorrelation, where observations closer in space are more likely to have similar values. Hence, this report applies spatial lag, spatial error, and geographically weighted regression (GWR) models to account for spatial dependence and local variation. The goal is to define these spatial methods that would provide a better fit and accurate interpretation than OLS.

Methods

A Description of the Concept of Spatial Autocorrelation

To begin with the 1st Law of Geography, "Everything is related to everything else, but near things are more related than distant things" by Waldo Tobler (1970). This concept is same as spatial autocorrelation reflects the idea that data points closer together in space tend to be more similar.

To detect global spatial autocorrelation, Moran's I defined as:

$$I = \frac{\left(\frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}\,(X_i - \bar{X})(X_j - \bar{X})}{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}}\right)}{\left(\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n}\right)} =$$

$$= \frac{n}{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}} \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}\,(X_i - \bar{X})(X_j - \bar{X})}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

where

- $\bar{X}$ is the mean of the variable X
- X_i is the variable value at a particular location *i*
- X_j is the variable value at another location *j*
- W_{ij} is a weight indexing location of *i* relative to *j*
- n is the number of observations (points or areal units)

(Kang, 2025, Lecture 4, slide 15)

The interpretation of Moran's I:

- I>0: Positive spatial autocorrelation, stated similar values cluster together.
- I<0: Negative spatial autocorrelation, stated dissimilar values are near each other (dispersion).
- I≈0: No spatial autocorrelation, stated random spatial pattern.

This report applies the Queen contiguity weight matrix, where a neighboring polygon shares either an edge or a corner. This matrix effectively captures spatial structure for irregular areal

units, such as census tracts. Statisticians often used more than one spatial weight matrix in their analyses, such as Rock and Queen, to check the robustness of their results.

A permutation test checks whether Moran's I is statistically significant. The null hypothesis (H0) states that the variable is spatially random. The alternative (H1) states that there is a significant spatial clustering. The permutation process involves randomly rearranging values across locations and recalculating Moran's I to build a reference distribution.

Local spatial autocorrelation identifies specific clusters (e.g., High-High or Low-Low areas). Unlike global Moran's I, local Indicators of Spatial Association (LISA) can highlight hotspots or outliers within a city.

A Review of OLS Regression and Assumptions

Ordinary Least Squares (OLS) regression is a foundational method for modeling relationships between a dependent variable and one or more independent variables. This model should be based on five assumptions to build on, such as a linear relationship between dependent and independent variables, residuals are normally distributed, residuals are random, residuals are homoscedastic, and independence of observations and residuals.

OLS assumptions often fail with spatial data due to spatial autocorrelation. To test for this, a common approach is to calculate Moran's I for the residuals. A significantly positive Moran's I suggests clustering of residuals, indicating a violation of the independence assumption.

Another method involves creating a scatterplot of OLS_RESIDU vs. WT_RESIDU, where WT_RESIDU represents spatially lagged residuals from neighboring units (defined using a

Queen contiguity matrix). The slope of this regression line is denoted by ρ (rho)—called lambda (λ) in GeoDa—and shown as Slope b in the output. A significant and positive ρ confirms spatial autocorrelation in the residuals.

GeoDa also provides tools to evaluate other regression assumptions, including testing for homoscedasticity and normality of errors. For testing for homoscedasticity, the assumption is that residuals have constant variance across observations. GeoDa tests for heteroscedasticity using the Breusch-Pagan test(BP), the Koenker-Bassett test (KB), and White's test. For the normality of errors, GeoDa tests this using the Jarque–Bera (JB) test. The hypothesis of both assumptions states below.

Hypotheses of Homoscedasticity:

- Null (H0): Residuals are homoscedastic (constant variance)
- Alternative (H1): Residuals are heteroscedastic (non-constant variance)

A significant result ($p < 0.05$) suggests violation of this assumption.

Hypotheses of Normality of Errors:

- Null (H0): Residuals are normally distributed
- Alternative (H1): Residuals are not normally distributed

A significant JB test result indicates that residuals deviate from normality, affecting the validity of inference (e.g., p-values, confidence intervals).

Spatial Lag and Spatial Error Regression

All spatial regressions in this section, including spatial lag and spatial error models, were conducted using GeoDa. Spatial lag regression assumes that the dependent variable in one

location is influenced by the same variable in nearby locations. It adds a spatially lagged dependent variable as a predictor to account for spatial spillover effects.

Spatial Lag Regression Model:

$$gini2i = \rho \sum wij \cdot gini2j + \beta 1 \cdot bachelori + \beta 2 \cdot withpubassi + \beta 3 \cdot lessthan10i + \beta 4 \cdot householdsi + \beta 5 \cdot femaleunemi + \varepsilon i$$

where,

- ρ: spatial autoregressive coefficient (indicates strength of spatial dependence),
- wij: spatial weight matrix (Queen contiguity),
- β1,…,β5: coefficients for the independent variables,
- εi: error term.

Continually, spatial error regression assumes that the spatial dependence lies in the error terms, not in the dependent variable directly. It accounts for omitted spatial variables that lead to autocorrelated errors.

Spatial Error Regression Model:

$$gini2i = \beta 1 \cdot bachelori + \beta 2 \cdot withpubassi + \beta 3 \cdot lessthan10i + \beta 4 \cdot householdsi + \beta 5 \cdot femaleunemi + \varepsilon i$$

$$\varepsilon i = \lambda \sum wij \cdot \varepsilon j + ui$$

where,

- β1,β2,…,β5: Coefficients for the explanatory variables,
- εi: The error term for observation i, which includes spatial autocorrelation,
- λ: Spatial autoregressive coefficient for the error term, indicating the degree of spatial dependence in the residuals,

- wij: Spatial weight between locations i and j, based on the Queen contiguity matrix,
- εj: The error term for neighboring observation j,
- ui: Random error term assumed to be i.i.d. with constant variance.

Both spatial lag and spatial error models still rely on most OLS assumptions, such as linearity, homoscedasticity, and normality of errors. However, they relax the assumption of independence of observations by directly modeling spatial autocorrelation.

The goal of using spatial lag and error models is to reduce spatial autocorrelation in the residuals, which could produce more accurate and unbiased coefficient estimates than those obtained with OLS.

To evaluate model performance, the following metrics are used:

1. Akaike Information Criterion (AIC) / Schwarz Criterion (SC or BIC)
    - Lower values indicate better model fit after penalizing for model complexity.
2. Log Likelihood (LL)
    - Higher values suggest a better fit to the data.
3. Likelihood Ratio Test (LRT)
    - Compares the spatial model against the nested OLS model.
    - Null Hypothesis (H0): OLS is sufficient (no spatial effect).
    - Alternative (H1): Spatial model improves fit.
    - A significant result supports using the spatial model.

Additionally, residual spatial autocorrelation is assessed using Moran's I. A model with lower or no spatial autocorrelation in the residuals is considered superior.

Geographically Weighted Regression (GWR)

In this study, the GWR analysis uses in ArcGIS Pro, which GWR models and visualizes local spatial variations in regression coefficients. The concept of GWR addresses Simpson's paradox, where global trends may mask or reverse local patterns. It applies local regression, meaning the relationship between variables is allowed to vary over space, rather than assuming constant effects across all locations as in OLS.

GWR Model: $yi = \beta 0(ui, vi) + \sum \beta k(ui, vi) xik + \varepsilon i$

Where:

- (ui,vi): coordinates of location i,
- βk(ui,vi): local coefficients estimated at location i,
- xik: value of predictor k at location i,
- εi: error term.

This means that each location gets its own unique regression equation.

Local regression is run at each target location, where closer observations are given more weight in the regression. The bandwidth determines the size of the neighborhood around each observation. ArcGIS selects it using a method like cross-validation, either with a fixed distance or adaptive (varies with point density).

GWR is still under some basic OLS assumptions. However, multicollinearity can become a serious issue locally. The Condition Number is used to detect it, if values over 30 may indicate problematic collinearity.

The p-value are not reported because each location has its own set of coefficients. The traditional p-values are not computed for GWR outputs. Instead, model fit is evaluated using diagnostics like AIC and local R-squared results.

This report focuses on global diagnostics, such as Akaike Information Criterion (AIC) for model comparison and Moran's I of GWR residuals (from GeoDa) to test for spatial autocorrelation. Local regression results are not discussed in detail, besides the local R-squared results.

Multiscale Geographically Weighted Regression (MGWR) extends GWR by allowing each predictor to operate at its own spatial scale. Unlike GWR, which uses one bandwidth for all variables, MGWR optimizes bandwidths separately, improving flexibility and accuracy.

Results

Spatial Autocorrelation

To evaluate global spatial autocorrelation of the Gini index (gini2). The calculation was conducted in GeoDa with 999 permutations. The result presents the Moran's I equals to 0.434 and Presudo p-value less than 0.001. This indicates that gini2 is significantly spatially autocorrelated, meaning similar Gini values tend to cluster geographically rather than being randomly distributed.

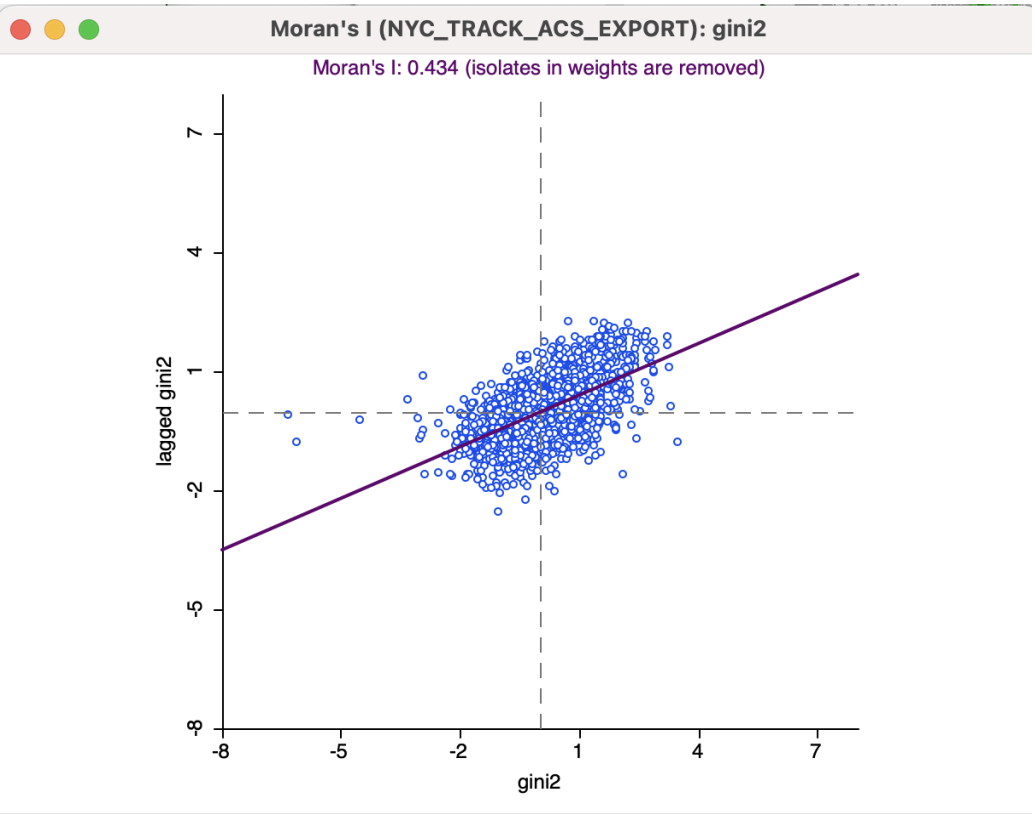


Figure 1. Global Moran's I Scatterplot for Gini Index (gini2)

The Local Moran's I (LISA) analysis to identify spatial clusters. The Significance Map and Cluster Map provide visual insights into where statistically significant spatial patterns occur.

- Not Significant (grey): 1407 tracts, no clear spatial pattern.

- High-High (red): 274 tracts, high Gini values surrounded by high values; these clusters are largely in Manhattan and parts of Brooklyn.
- Low-Low (blue): 324 tracts — low Gini values surrounded by low values; mostly in southern Queens and Staten Island.
- High-Low (pink): 64 tracts — high values surrounded by low; scattered across Brooklyn and Bronx.
- Low-High (light blue): 41 tracts — low values surrounded by high; small clusters in central Brooklyn.

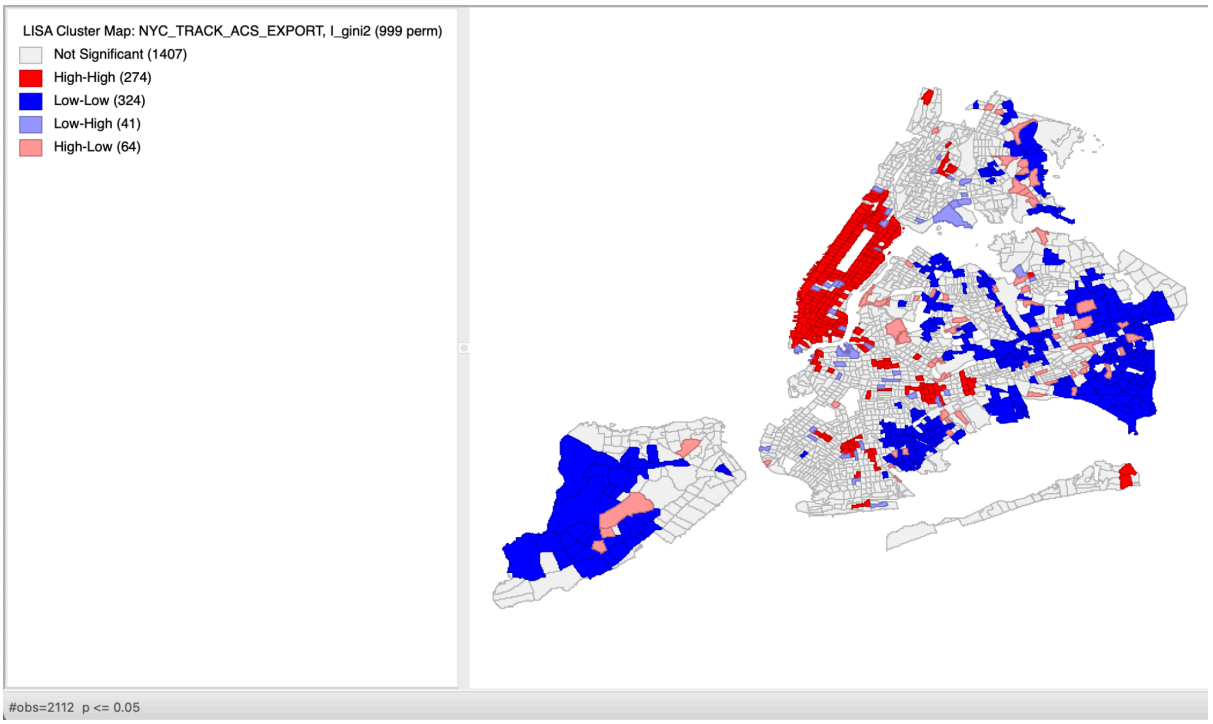


Figure 2. LISA Cluster Map of Gini Index (gini2)

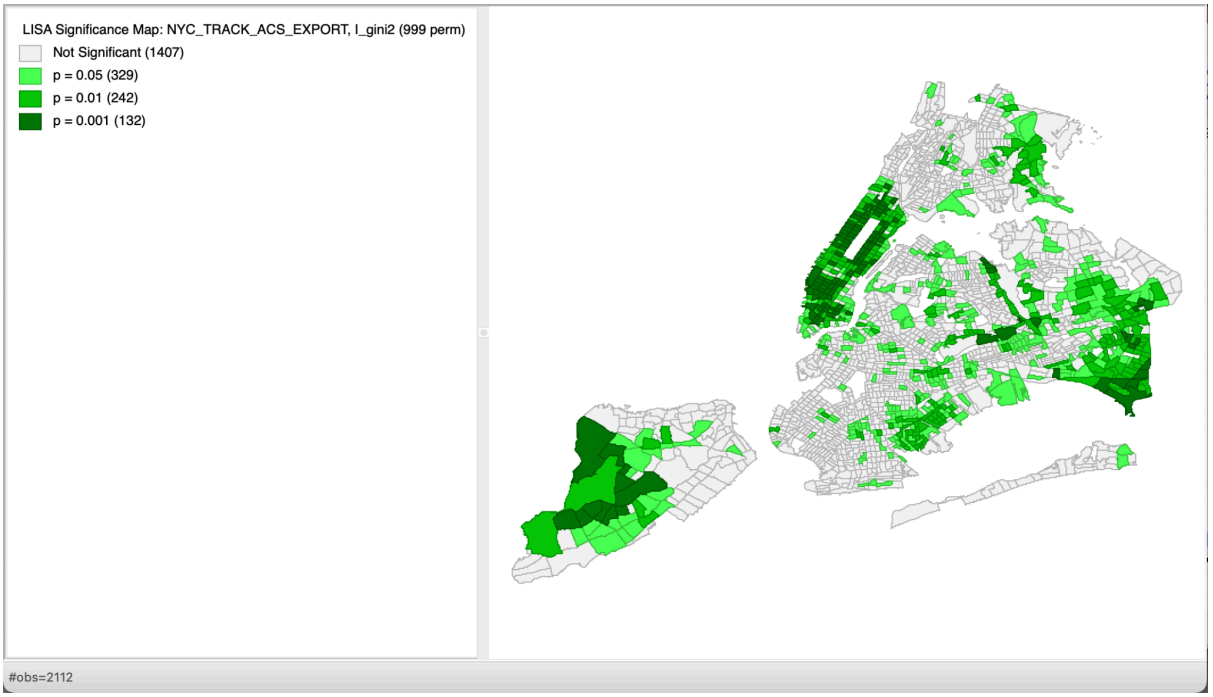


Figure 3. LISA Significance Map (p-values) for Gini Index

A Review of OLS Regression and Assumptions: Results

```
REGRESSION
----------
SUMMARY OF OUTPUT: ORDINARY LEAST SQUARES ESTIMATION
Data set            :  NYC_TRACK_ACS_EXPORT
Dependent Variable  :       gini2  Number of Observations: 2112
Mean dependent var  :    0.443775  Number of Variables   :    6
S.D. dependent var  :   0.0668558  Degrees of Freedom    : 2106

R-squared           :    0.306652  F-statistic           :     186.287
Adjusted R-squared  :    0.305006  Prob(F-statistic)     :           0
Sum squared residual:     6.54521  Log likelihood        :     3103.35
Sigma-square        :  0.00310788  Akaike info criterion :    -6194.71
S.E. of regression  :   0.0557484  Schwarz criterion     :    -6160.77
Sigma-square ML     :  0.00309906
S.E of regression ML:   0.0556692

------------------------------------------------------------------------------
       Variable      Coefficient      Std.Error    t-Statistic   Probability
------------------------------------------------------------------------------
          CONSTANT        0.4185     0.00229834        182.088     0.00000
          bachelor      7.21e-05    5.68544e-06        12.6815     0.00000
        withpubass   7.35664e-05    2.89953e-05        2.53719     0.01125
        lessthan10   0.000282612    1.42718e-05        19.8021     0.00000
        households  -3.37995e-05     3.8723e-06       -8.72853     0.00000
        femaleunem  -0.000119374    2.04696e-05       -5.83176     0.00000
------------------------------------------------------------------------------

REGRESSION DIAGNOSTICS
MULTICOLLINEARITY CONDITION NUMBER   14.390986
TEST ON NORMALITY OF ERRORS
TEST                   DF           VALUE             PROB
Jarque-Bera             2           963.9031          0.00000

DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST                   DF           VALUE             PROB
Breusch-Pagan test      5            61.8008          0.00000
Koenker-Bassett test    5            23.2789          0.00030
SPECIFICATION ROBUST TEST
TEST                   DF           VALUE             PROB
White                  20           113.4962          0.00000

DIAGNOSTICS FOR SPATIAL DEPENDENCE
FOR WEIGHT MATRIX : NYC_TRACK_ACS_EXPORT
   (row-standardized weights)
TEST                          MI/DF        VALUE          PROB
Moran's I (error)             0.2525       19.7006        0.00000
Lagrange Multiplier (lag)       1         468.7911        0.00000
Robust LM (lag)                 1         101.4268        0.00000
Lagrange Multiplier (error)     1         381.3142        0.00000
Robust LM (error)               1          13.9500        0.00019
Lagrange Multiplier (SARMA)     2         482.7410        0.00000
```

Table 1. OLS Regression Results for Gini Index (gini2) in NYC

The OLS regression model with predictors (bachelor, withpubass, lessthan10, households, and femaleunem) explains 30.5% of the variation in the Gini index(Adjusted R squared = 0.3050), and the statistically significant with p-values < 0.05.

The results of 3 tests (Breusch–Pagan, Koenker–Bassett, and White test) on heteroscedasticity are consistent. All of the tests have p-values < 0.001, indicating heteroscedasticity is present in the model, which violates the OLS assumption of constant variance. Furthermore, the Jarque–Bera test reports a value of 963.90 with a p-value = 0.00000, indicating that the residuals are not normally distributed. These tests show that the model is not under assumptions of constant variance and normally distributed.

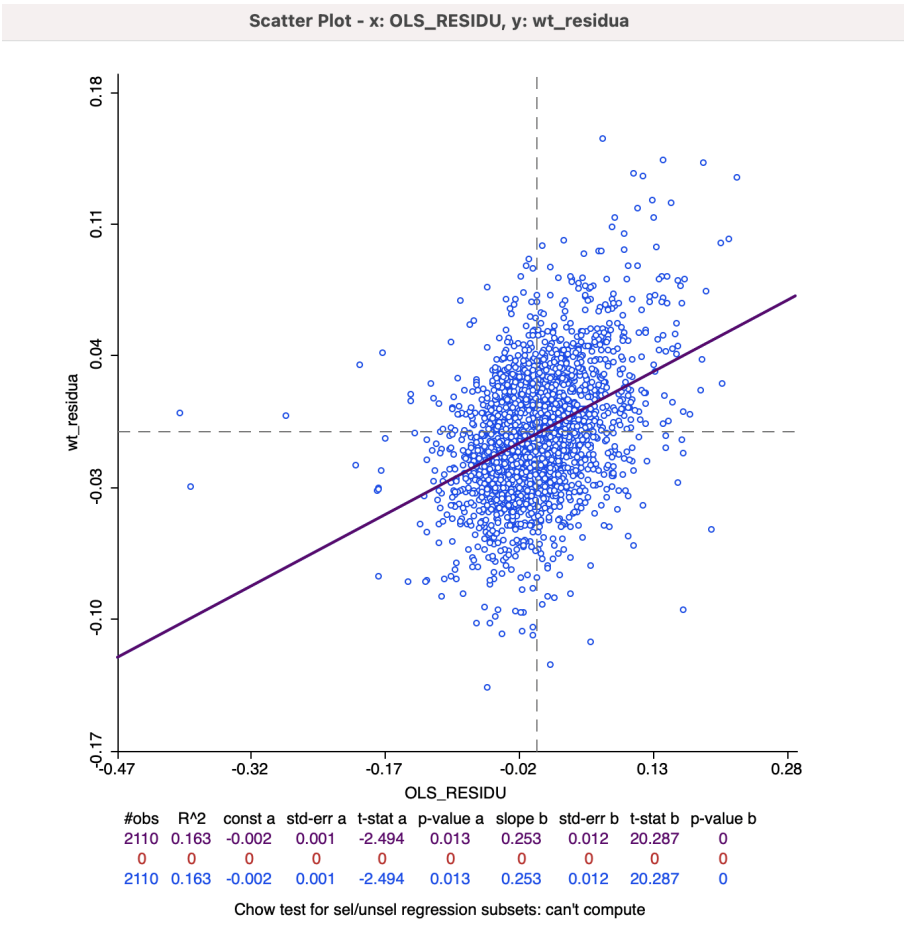


Figure 4. Scatter Plot of OLS Residuals (OLS_RESIDU) vs. Spatially Lagged Residuals (WT_RESIDU)

The Figure 4 presents the scatterplot of OLS_RESIDU versus WT_RESIDU, which visualizes the spatial lag of residuals. The slope coefficient (ρ) is 0.253 with a p-value of 0.000, indicating statistically significant spatial autocorrelation. This suggests that residuals are spatially dependent, violating the OLS assumption of independently distributed errors.

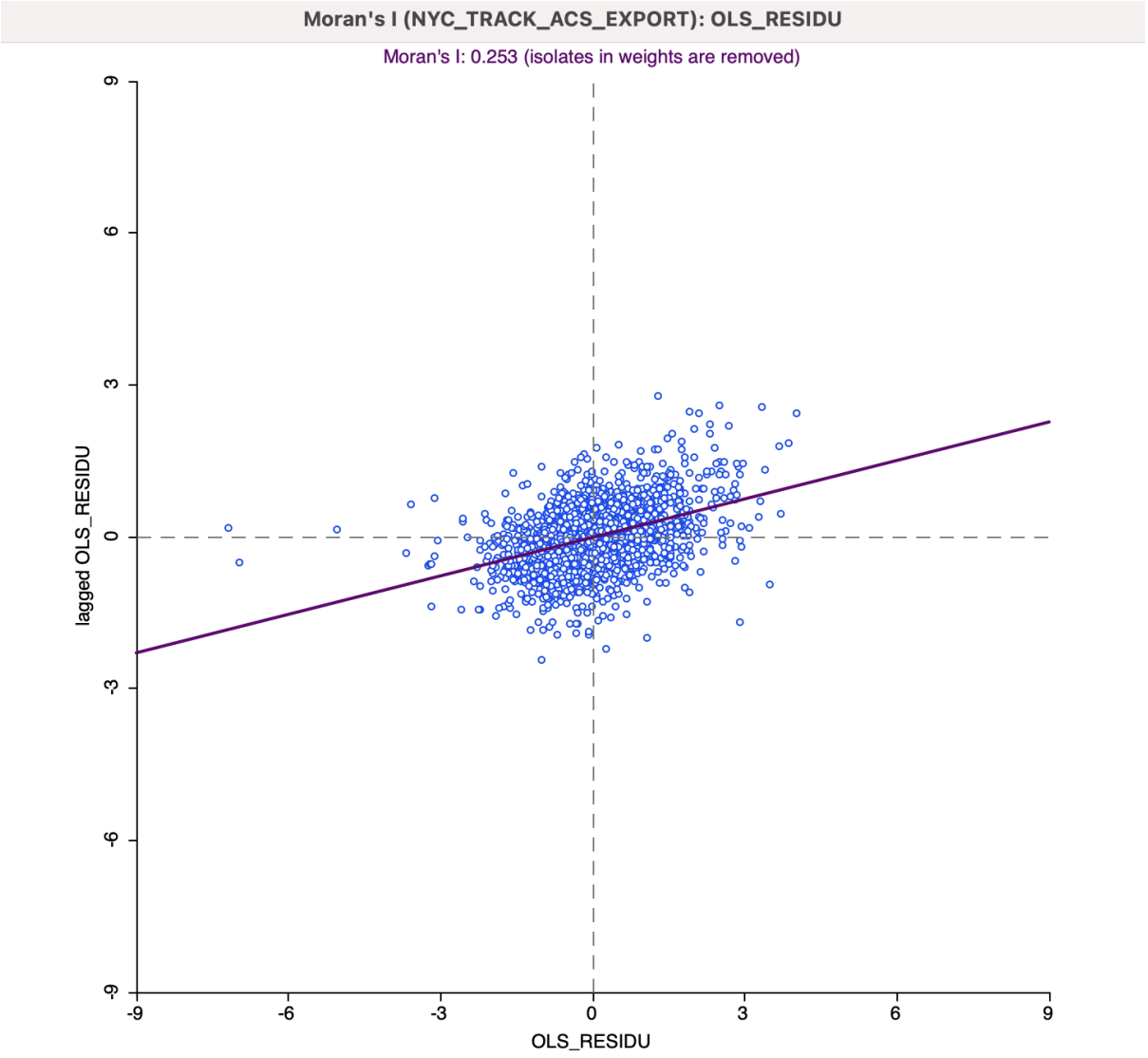


Figure 5. Moran's I Scatterplot for OLS Residuals

The scatterplot of OLS residuals show significant spatial autocorrelation, as indicated by a Moran's I value of 0.253 with a p-value of 0.000 (based on 999 permutations). This is problematic, which violates the OLS assumption of spatial independence in the residuals. This causes spatial effects are not adequately captured by the OLS model, and suggests to do

alternative models, such as spatial lag or spatial error regression should be used to better account for the spatial structure in the data.

Spatial Lag and Spatial Error Regression Results

Spatial Lag Results

```
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REGRESSION
----------
SUMMARY OF OUTPUT: SPATIAL LAG MODEL - MAXIMUM LIKELIHOOD ESTIMATION
Data set            : NYC_TRACK_ACS_EXPORT
Spatial Weight      : NYC_TRACK_ACS_EXPORT
Dependent Variable  :       gini2  Number of Observations: 2110
Mean dependent var  :    0.443823  Number of Variables   :    7
S.D. dependent var  :   0.0668662  Degrees of Freedom    : 2103
Lag coeff.   (Rho)  :    0.452552

R-squared           :    0.439765  Log likelihood        :     3209.43
Sq. Correlation     : -            Akaike info criterion :    -6404.86
Sigma-square        :  0.00250486  Schwarz criterion     :    -6365.28
S.E of regression   :   0.0500486

------------------------------------------------------------------------
       Variable       Coefficient     Std.Error       z-value    Probability
------------------------------------------------------------------------
           W_gini2      0.452552      0.0235772        19.1945     0.00000
          CONSTANT      0.227129      0.0103126        22.0245     0.00000
          bachelor   4.55639e-05     5.2005e-06        8.76146     0.00000
        withpubass   3.48191e-05    2.60784e-05        1.33517     0.18182
        lessthan10   0.000218026    1.32621e-05        16.4397     0.00000
        households  -2.42454e-05    3.50657e-06       -6.91428     0.00000
        femaleunem  -8.08401e-05    1.84101e-05       -4.39108     0.00001
------------------------------------------------------------------------

REGRESSION DIAGNOSTICS
DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST                                    DF      VALUE        PROB
Breusch-Pagan test                      5        29.1768     0.00002

DIAGNOSTICS FOR SPATIAL DEPENDENCE
SPATIAL LAG DEPENDENCE FOR WEIGHT MATRIX : NYC_TRACK_ACS_EXPORT
TEST                                    DF      VALUE        PROB
Likelihood Ratio Test                   1       219.6530     0.00000
```

Table 2. Spatial Lag Regression Results for Gini Index

The table 2 presents the spatial lag coefficient, $\rho = 0.4526$, and it is highly significant, $p < 0.001$. This confirms strong spatial dependence in the dependent variable (gini2). In other words, a tract's income inequality is positively associated with the inequality of neighboring tracts, validating the use of a spatial lag model. For the predictors, the majority of them are significant at the 0.05 level, besides withpubass is not significant, $p = 0.18182$. The

comparison to OLS, coefficients and significance are generally consistent, but some effect sizes slightly shift due to the model accounting for spatial dependence.

The Breusch-Pagan test returns with value = 29.1768, p = 0.00002, indicating significant heteroskedasticity still exists in the spatial lag model. Therefore, the assumption of constant error variance is violated.

Below are the model fit comparison between spatial lag and OLS,

- AIC: Spatial Lag = -6404.86 vs OLS = ~-6194.71 → Spatial Lag is better.
- Schwarz Criterion: Spatial Lag = -6365.28 vs OLS = ~-6160.77 → Spatial Lag is better.
- Log Likelihood: Spatial Lag = 3209.43 vs OLS = 3103.35 → Spatial Lag is better.
- Likelihood Ratio Test: LR = 219.6530, p < 0.0001 → Spatial Lag significantly improves fit over OLS.

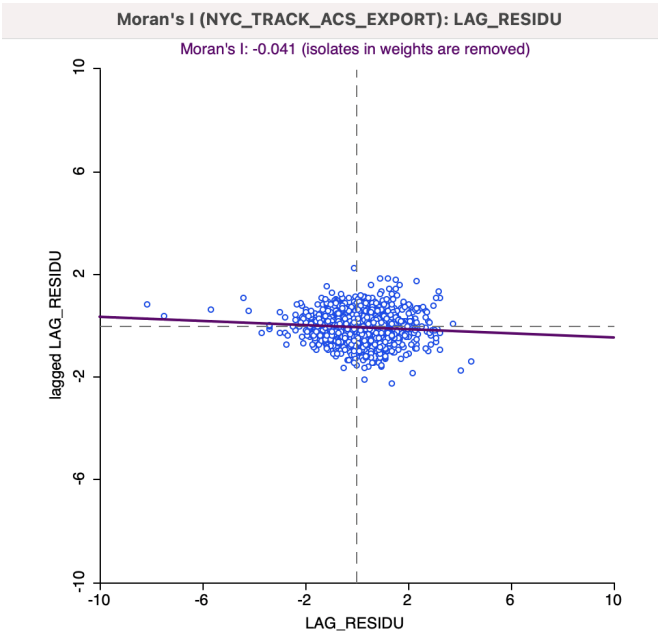


Figure 6. Moran's I Scatterplot of Residuals from Spatial Lag Model

Figure 6 shows the Moran's I scatterplot of the spatial lag model residuals, which has a Moran's I of -0.041, compared to 0.253 in the OLS residuals. This reduction indicates that the spatial lag model effectively addressed the spatial autocorrelation problem, confirming that spatial dependence in the original model has been corrected.

The Spatial Lag model performs better than the OLS model in terms of model fit and accounting for spatial dependence. However, the assumption of heteroskedasticity is still an issue in both models.

Spatial Error

```
REGRESSION
----------
SUMMARY OF OUTPUT: SPATIAL ERROR MODEL - MAXIMUM LIKELIHOOD ESTIMATION
Data set            : NYC_TRACK_ACS_EXPORT
Spatial Weight      : NYC_TRACK_ACS_EXPORT
Dependent Variable  :       gini2  Number of Observations: 2110
Mean dependent var  :    0.443823  Number of Variables   :    6
S.D. dependent var  :    0.066866  Degrees of Freedom    : 2104
Lag coeff. (Lambda) :    0.533665

R-squared           :    0.434758  R-squared (BUSE)      : -
Sq. Correlation     : -            Log likelihood        : 3184.624529
Sigma-square        :  0.00252725  Akaike info criterion :    -6357.25
S.E of regression   :   0.0502718  Schwarz criterion     :    -6323.32

------------------------------------------------------------------------------
       Variable       Coefficient     Std.Error       z-value    Probability
------------------------------------------------------------------------------
          CONSTANT      0.428898     0.00310433        138.161     0.00000
          bachelor   4.12108e-05    5.97178e-06        6.90093     0.00000
        withpubass    3.2363e-05    2.73495e-05        1.18331     0.23669
        lessthan10   0.000232743    1.40645e-05        16.5482     0.00000
        households  -2.53457e-05     3.7799e-06       -6.7054     0.00000
        femaleunem  -6.00067e-05    1.98302e-05       -3.02603     0.00248
            LAMBDA      0.533665       0.025274        21.1152     0.00000
------------------------------------------------------------------------------

REGRESSION DIAGNOSTICS
DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST                                    DF      VALUE        PROB
Breusch-Pagan test                      5        21.7884     0.00057

DIAGNOSTICS FOR SPATIAL DEPENDENCE
SPATIAL ERROR DEPENDENCE FOR WEIGHT MATRIX : NYC_TRACK_ACS_EXPORT
TEST                                    DF      VALUE        PROB
Likelihood Ratio Test                   1       170.0440     0.00000
```

Table 3. Spatial Error Regression Results for Gini Index (gini2)

The table 3 presents the spatial error coefficient , $\lambda = 0.5337$, and it is highly significant, $p < 0.001$. This confirms spatial error dependence exists, which means unobserved spatial processes are influencing the dependent variable (Gini index), and the spatial error model accounts for that spatial structure in the residuals. For the predictors, the majority of them are significant at the 0.05 level, besides withpubass is not significant, $p = 0.23669$. The results are mostly consistent with OLS and spatial lag outputs, though coefficients and z-values are adjusted due to error structure correction.

The Breusch-Pagan test yields a value of 21.7884, $p = 0.00057$, indicating significant heteroskedasticity still exists in the spatial error model. Therefore, the assumption of constant error variance is violated.

Below are the model fit comparison between spatial error and OLS,

- AIC: Spatial Error = -6357.25 vs OLS = ~-6194.71 → Spatial Error is better.
- Schwarz Criterion: Spatial Error = -6323.32 vs OLS = ~-6160.77 → Spatial Error is better.
- Log Likelihood: Spatial Error = 3184.62 vs OLS = 3103.35 → Spatial Error is better.
- Likelihood Ratio Test: LR = 170.0440, $p < 0.0001$ → Spatial Error significantly improves fit over OLS.

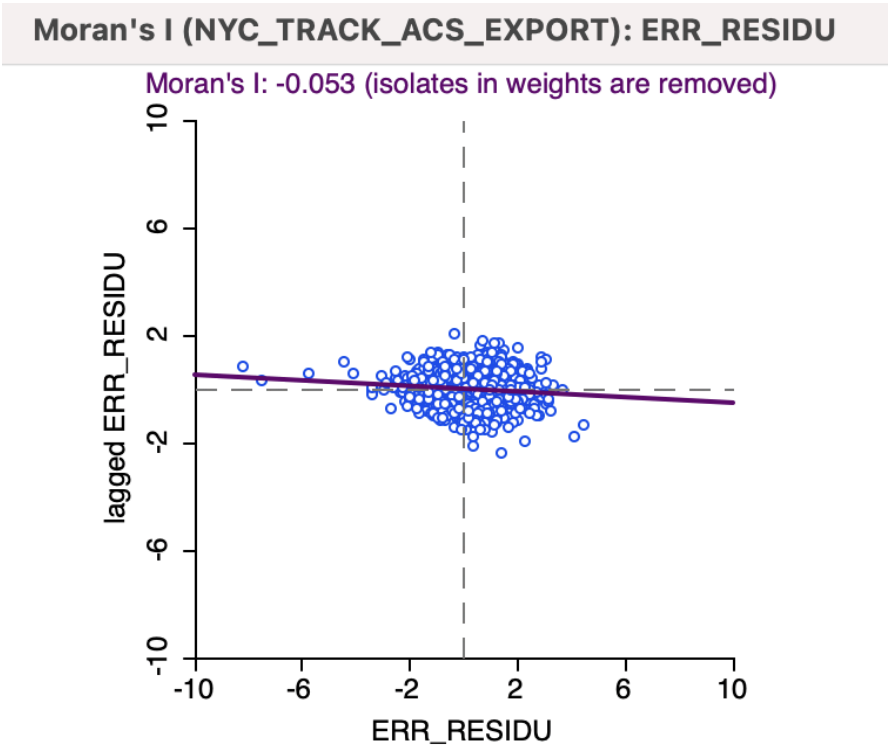


Figure 7. Moran's I Scatterplot of Spatial Error Model Residuals

Figure 7 shows the Moran's I scatterplot of the spatial error model residuals, which has a Moran's I of -0.053, compared to 0.253 in the OLS residuals. This reduction indicates that the spatial error model effectively addressed the spatial autocorrelation problem, confirming that spatial dependence in the residuals has been addressed.

Based on all model comparison criteria, both the spatial lag and spatial error models outperform the OLS model. They exhibit lower AIC and Schwarz Criterion values, higher log-likelihoods, and significantly reduce the spatial autocorrelation in the residuals. Between the two, the spatial lag model shows slightly stronger performance, particularly in terms of R squared (0.4398 vs. 0.4348) and a more substantial reduction in residual spatial dependence (Moran's I = -0.041 vs. -0.053). Therefore, the spatial lag model is considered the best-fitting model for explaining the spatial pattern of income inequality in this analysis.

The spatial error model has better performance than OLS across all key metrics. It has a lower AIC and Schwarz Criterion, higher log-likelihood, and significantly reduced spatial autocorrelation in residuals (Moran's I: -0.053 vs. 0.253). Thus, spatial error is the better model.

When comparing the spatial lag and spatial error models, the spatial lag model has better performance based on model fit criteria. Specifically, the Akaike Information Criterion (AIC) for the spatial lag model is -6404.86, compared to -6357.25 for the spatial error model. Similarly, the Schwarz Criterion (SC) is -6365.28 for the spatial lag model and -6323.32 for the spatial error model. Since both AIC and SC values are lower in the spatial lag model, it indicates a better overall fit, making the spatial lag model the preferred specification for capturing spatial dependence in the data.

Geographically Weighted Regression Results

Model Diagnostics

R2	0.4217
AdjR2	0.4046
AICc	-6493.1895
Sigma-Squared	0.0027
Sigma-Squared MLE	0.0026
Effective Degrees of Freedom	2051.3676
Adjusted Critical Value of Pseudo-t Statistics	2.7243

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Table 4. Model Diagnostics from Geographically Weighted Regression (GWR)

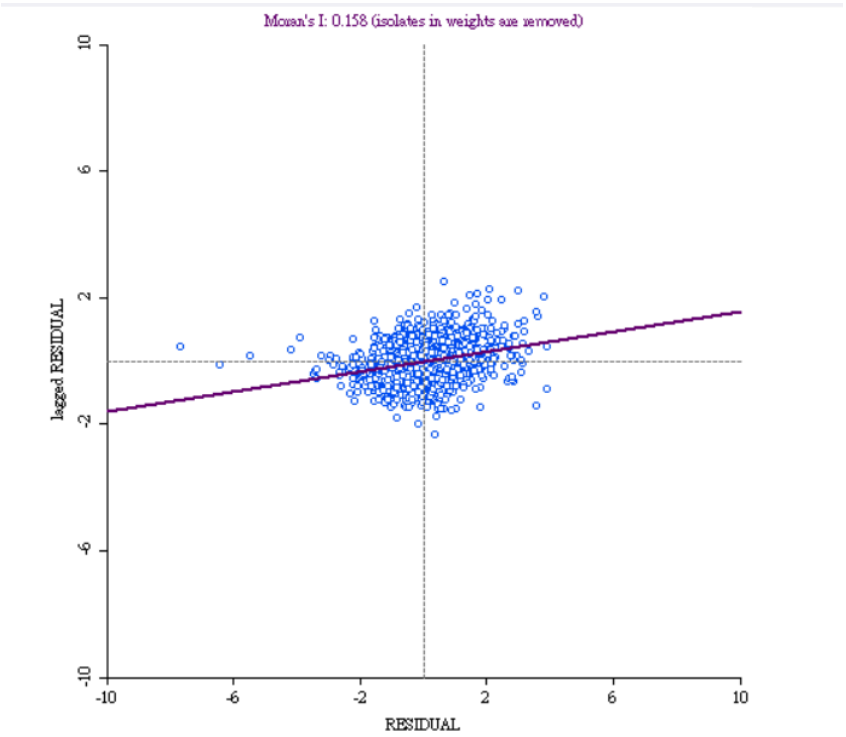


Figure 8. Moran's I Scatterplot of GWR Residuals

Figure 8 presents the Moran's I scatterplot of the GWR residuals, with a Moran's I value of 0.158. This value is lower than the OLS residuals' Moran's I of 0.253, suggesting that the GWR model has reduced spatial autocorrelation to some extent. However, it is still higher than that of the spatial lag (−0.041) and spatial error (−0.053) models, which effectively removed most spatial dependence in the residuals. Thus, while GWR improves over OLS in addressing spatial autocorrelation, it is less effective than the spatial lag and spatial error models in this regard.

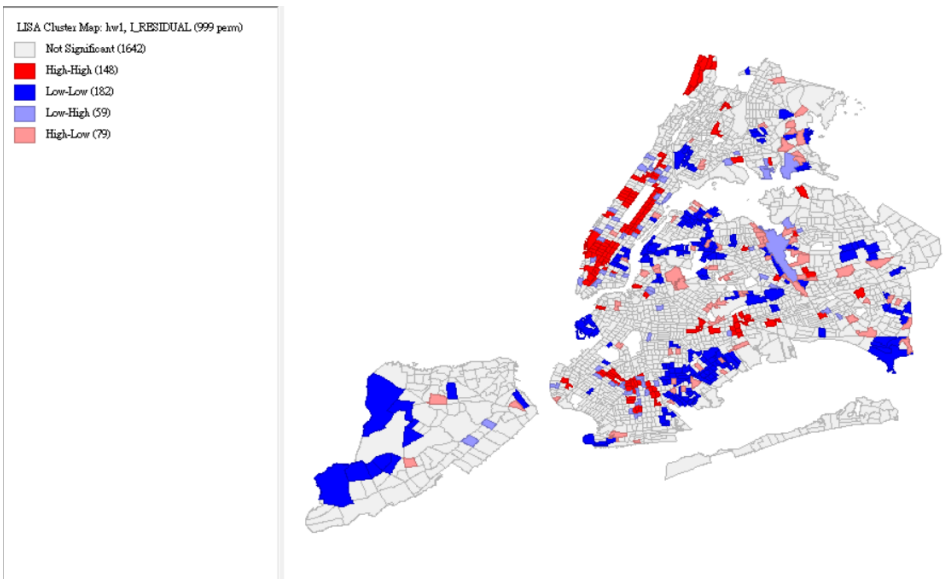


Figure 9. LISA Cluster Map of GWR Residuals

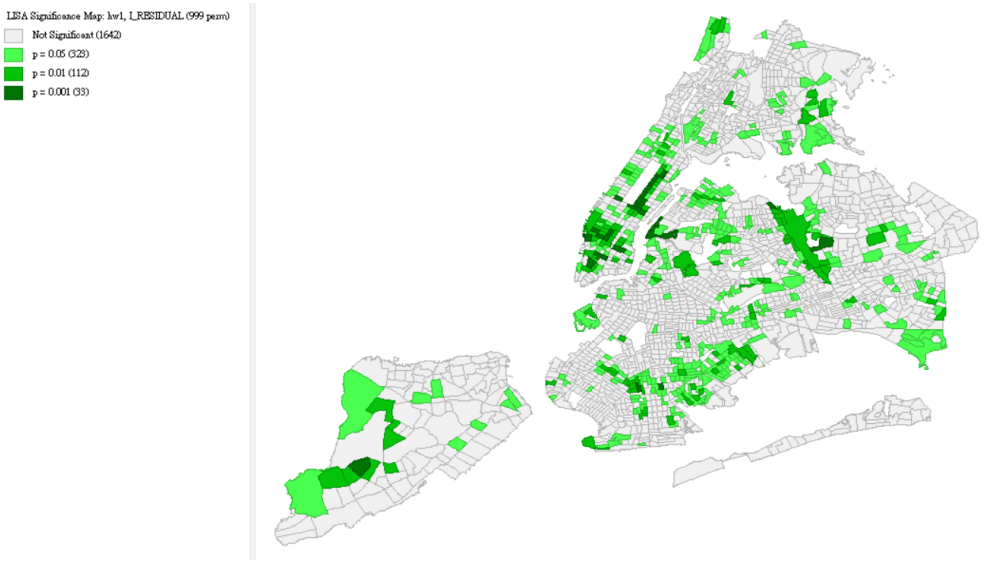


Figure 10. LISA Significance Map of GWR Residuals (p-values)

The GWR results show spatial variation in how predictors explain income inequality across NYC. High R squared values in southern Brooklyn and Staten Island suggest stronger local fits, while areas like Manhattan show weaker performance. The LISA map reveals some significant clusters, confirming that GWR captures local patterns better than OLS.

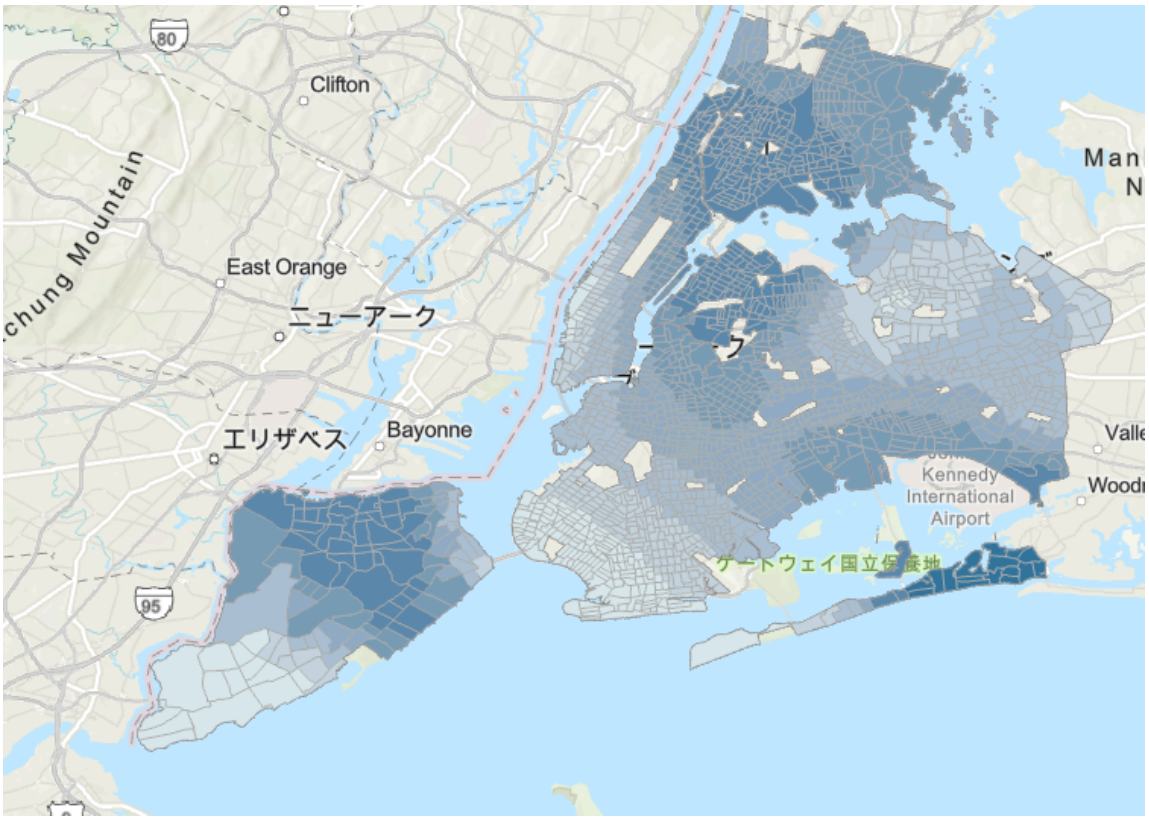


Figure 11. Choropleth Map of Local R-squared Values from GWR Model

The Figure 11 presents the choropleth map displays the local R-squared values from the Geographically Weighted Regression (GWR) model. Areas with darker shades have higher R squared values, meaning the model explains more variance in the Gini index there. Higher R squared is observed in southern Brooklyn and parts of Staten Island, suggesting stronger local relationships between the predictors and income inequality. On the other hand, some central and northern parts of NYC have lower R squared values, indicating weaker model fit in those areas. This spatial variation highlights the benefit of using GWR to capture local differences in regression strength.

Multiscale Geographically Weighted Regression (MGWR)

Model Diagnostics

Statistic	GWR	MGWR
R-Squared	0.4217	0.4284
Adjusted R-Squared	0.4046	0.4169
AICc	4933.4397	4899.6452
Sigma-Squared	0.5954	0.5831
Sigma-Squared MLE	0.5783	0.5716
Effective Degrees of Freedom	2051.3658	2070.1924

Optimal GWR Bandwidth: 11713.22 meters (Distance).

Table 5. Model Diagnostics from Multiscale Geographically Weighted Regression (MGWR)

The MGWR model shows slightly better performance than GWR across most diagnostics. From comparing the R squared, 0.4284 vs. 0.4217, this indicates more variance explained. The adjusted R squared is also higher, and the AICc is lower, 4899.65 vs. 4933.44, suggesting a better model fit. Additionally, MGWR has slightly lower sigma-squared values, meaning less residual variation. Overall, MGWR provides a more refined and accurate spatial model than standard GWR.

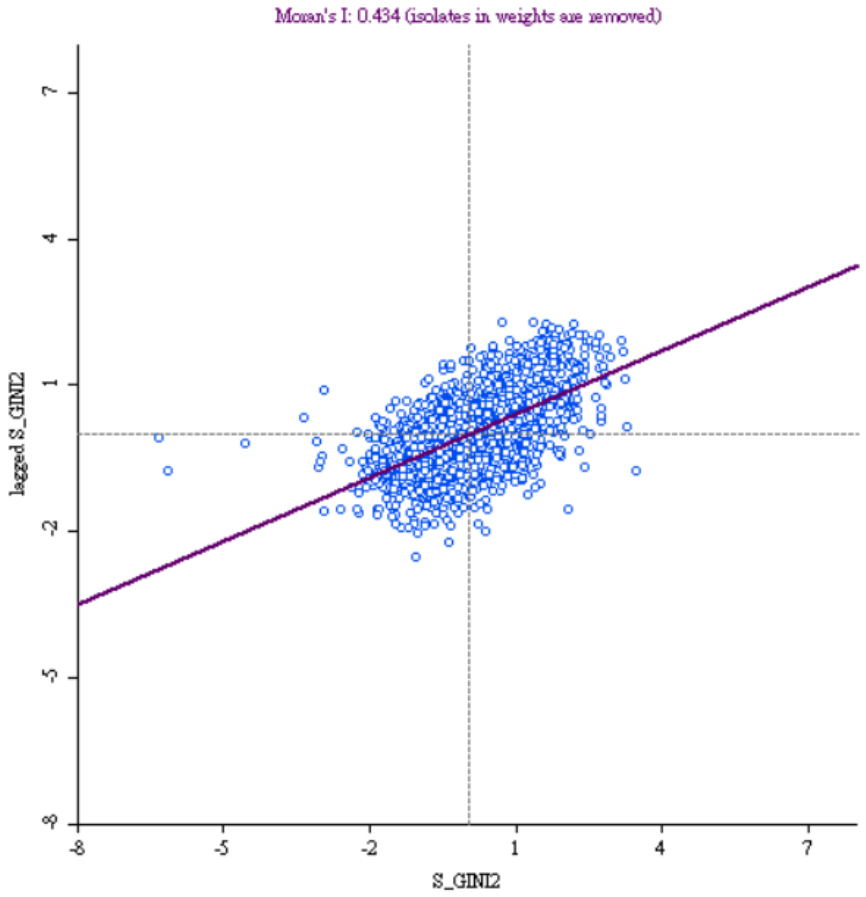


Figure 12. Moran's I Scatterplot of MGWR Residuals

The Moran's I of 0.434 shows moderate residual spatial autocorrelation in the MGWR model, meaning it improved fit but didn't fully remove spatial dependence.

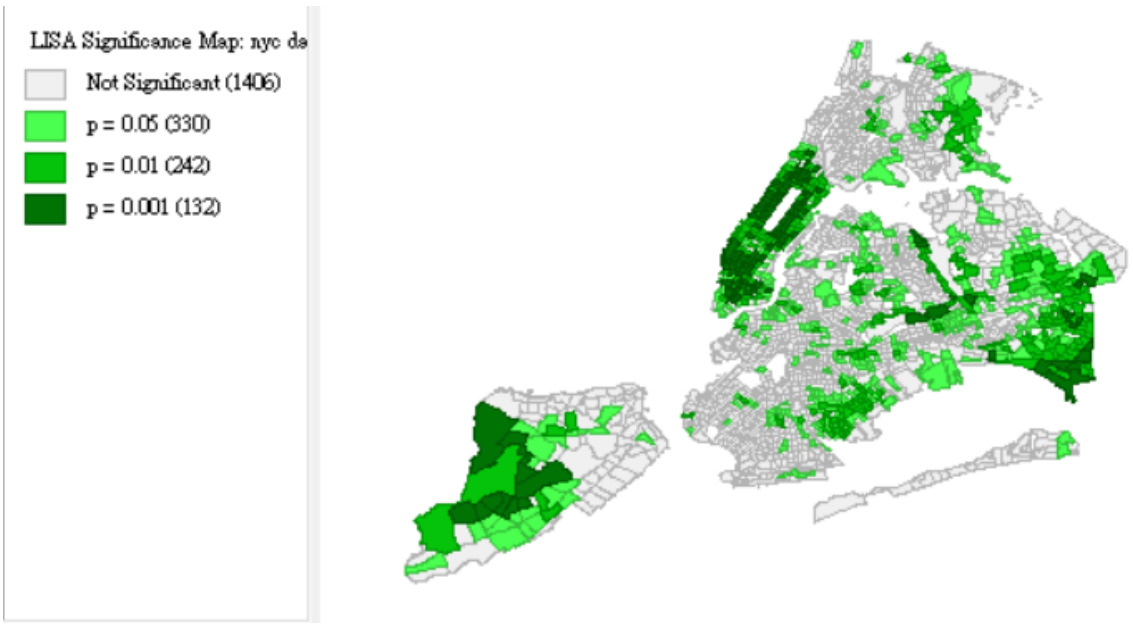


Figure 13. LISA Significance Map of MGWR Residuals

From Figure 13, it highlights where MGWR residuals are statistically significant. Areas shaded in green ($p < 0.05, 0.01, 0.001$) indicate tracts where residuals are significantly spatially clustered. This suggests that, although MGWR improves model fit, some spatial patterns in residuals still remain.

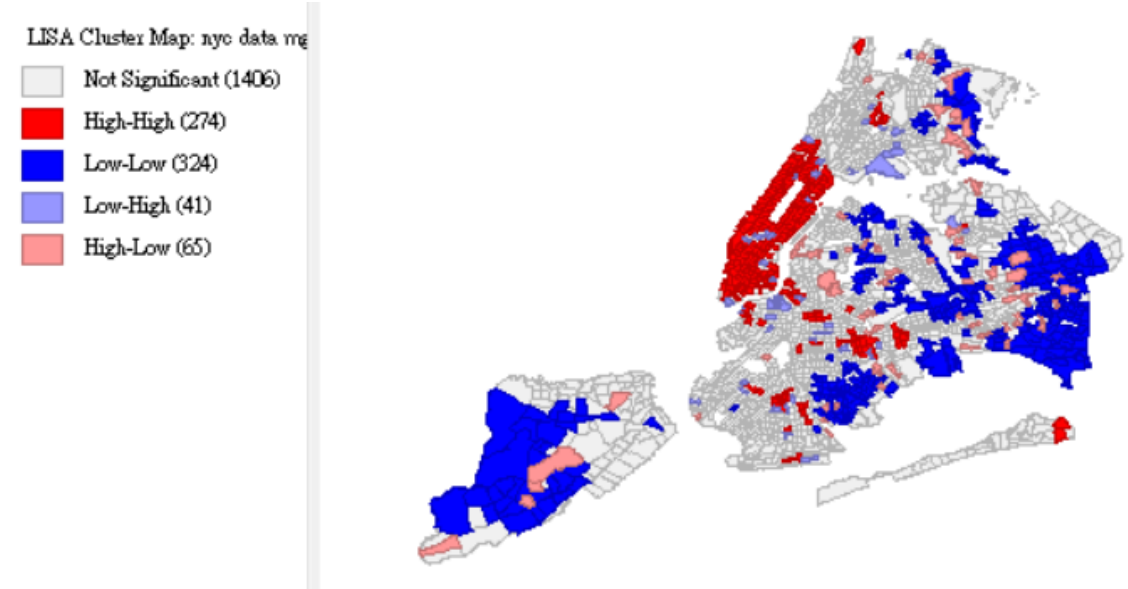


Figure 14. LISA Cluster Map of MGWR Residuals

From Figure 14, LISA Cluster Map shows spatial clustering of MGWR residuals. Red areas (High-High) and blue areas (Low-Low) indicate clustered over or under predictions, meaning some spatial autocorrelation remains even after using MGWR.

Discussion

This study analyzed the spatial distribution of income inequality in New York City using various regression models, including OLS, spatial lag, spatial error, GWR, and MGWR. While OLS showed significant relationships between the Gini index and predictors, it failed to account for spatial dependence. Spatial lag and spatial error models substantially improved model fit and reduced spatial autocorrelation in the residuals, with the spatial lag model performing best based on AIC and residual diagnostics. GWR and MGWR captured local variations and slightly improved the fit over OLS, but did not entirely remove spatial dependence. MGWR offered the highest R squared and lowest AIC among local models.

However, limitations exist. Across most models, heteroskedasticity was present (Breusch-Pagan tests were significant), and OLS residuals were not normally distributed (Jarque–Bera test). GWR and MGWR do not report standard p-values, and interpretation depends on spatial context. Some spatial autocorrelation persisted even after modeling, particularly in GWR and MGWR, suggesting that not all spatial processes were captured.

Income inequality is evident in New York City, as reflected in uneven patterns across neighborhoods. Certain areas exhibit higher concentrations of wealth, while others consistently face economic disadvantages. A combination of historical development, access to education and employment, and systemic social factors shapes these disparities. While statistical models can measure these patterns, the presence of inequality itself is clear and calls for thoughtful, place-based policy responses.